

Chem 273 Discussion 3

Reminders

- Lecture Exercise 1 Grades have been published
- You may submit re-grades on Gradescope
- Lecture Exercise 3 is due next Wednesday the 11th at 11:59PM

Lab Participation

You are expected to attend each Lab Session and participate

You don't have to unmute or answer technical questions, but please consider responding in the chat, asking questions, replying to your peers

Lecture Exercise 3

There is NO coding involved in this exercise

Feel free to submit any file type (i.e. pdf, word document, scanned in notes, etc.)

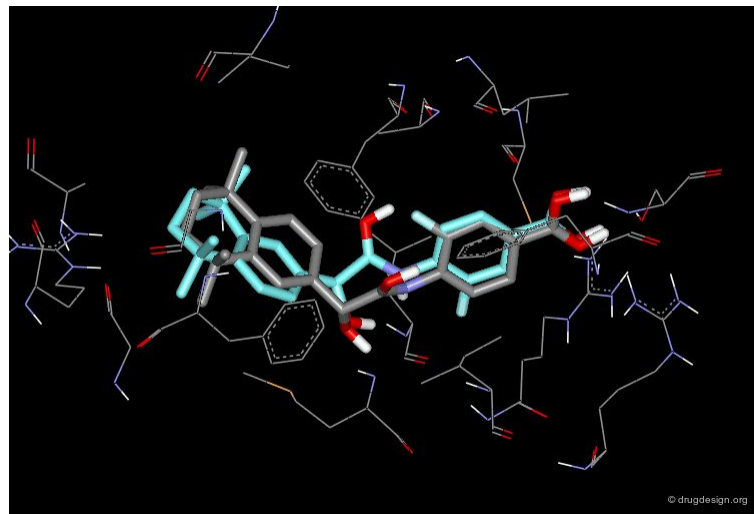
Lecture Exercise 2 solution

How matrices are important to drug discovery!

Molecules or Ligands (when bound to proteins) are represented in 3D space by coordinates (x, y, z).

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107.008000 15.860000 99.456000  
106.932000 16.930000 100.366000
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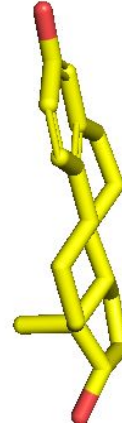
This is a matrix of shape (n_atoms, 3).
Applying operations on the matrix can
allows us to manipulate the molecule
structure!



Relevant for: protein-ligand docking,
generative drug design,
substructure searching, etc.



Matrix Operations



We can create many different poses for ligands!

Ligand translation demo

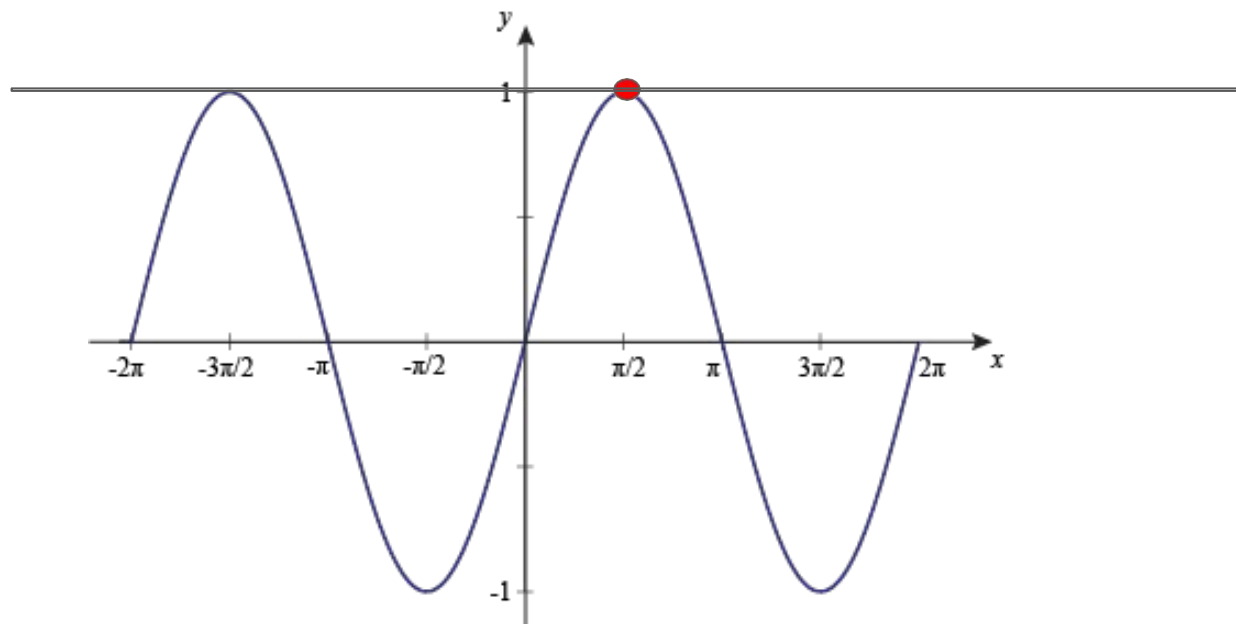
Derivative Practice

$$f(x) = 6x^3 - 9x + 4$$

$$f'(x) = 18x^2 - 9$$

Derivative Practice

Derivative at $\pi/2$?



Derivative Practice

$$g(z) = 4z^7 - 3z^{-7} + 9z$$

$$g'(z) = 28z^6 + 21z^{-8} + 9$$

Taylor Series

$$f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2$$